S20545_Programming_Assignment_01

March 1, 2026

1 Phase 1: Environment Setup and Exploratory Data Analysis (EDA)

Note: The test dataset is strictly isolated and will not be analyzed to prevent data leakage.

[4]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import torch
from collections import Counter
from wordcloud import WordCloud
import nltk
from nltk.corpus import stopwords

nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

sns.set_theme(style="whitegrid")

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Using device: {device}")
if device.type == 'cuda':
    print(f"GPU Name: {torch.cuda.get_device_name(0)}")
```

```
Using device: cuda
GPU Name: NVIDIA GeForce RTX 3050 Laptop GPU
```

```
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\Ravin\AppData\Roaming\nltk_data…
[nltk_data]   Package stopwords is already up-to-date!
```

1.1 1.1 Loading the Training Data and Basic Statistics

[5]:
```
df_train = pd.read_csv('phm_train.csv')

print("--- Dataset Info ---")
print(df_train.info())
```

```
print("\n--- First 5 Rows ---")
display(df_train.head())
```

```
--- Dataset Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9991 entries, 0 to 9990
Data columns (total 3 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   tweet_id  9991 non-null   float64
 1   label     9991 non-null   int64
 2   tweet     9991 non-null   object
dtypes: float64(1), int64(1), object(1)
memory usage: 234.3+ KB
None

--- First 5 Rows ---

       tweet_id  label                                              tweet
0  6.430000e+17      0  user_mention all i can tell you is i have had …
1  6.440000e+17      0  my doctor told me stop he gave me sum pop i mi…
2  8.150000e+17      1  i take tylenol and i wake up in the middle of …
3  6.820000e+17      0  i got xans in an advil bottle i dont take them…
4  6.440000e+17      1  mom says i need to stop eating so much bc ive …
```

[6]:
```
print("\n--- Missing Values ---")
print(df_train.isnull().sum())
```

```
--- Missing Values ---
tweet_id    0
label       0
tweet       0
dtype: int64
```

[7]:
```
plt.figure(figsize=(6, 4))
sns.countplot(data=df_train, x='label', hue='label', palette='viridis',␣
 ↪legend=False)
plt.title('Class Distribution in Training Data')
plt.xlabel('Label (0: Non-Personal, 1: Personal)')
plt.ylabel('Count')
plt.show()

print("\nClass Counts:")
print(df_train['label'].value_counts(normalize=True) * 100)
```

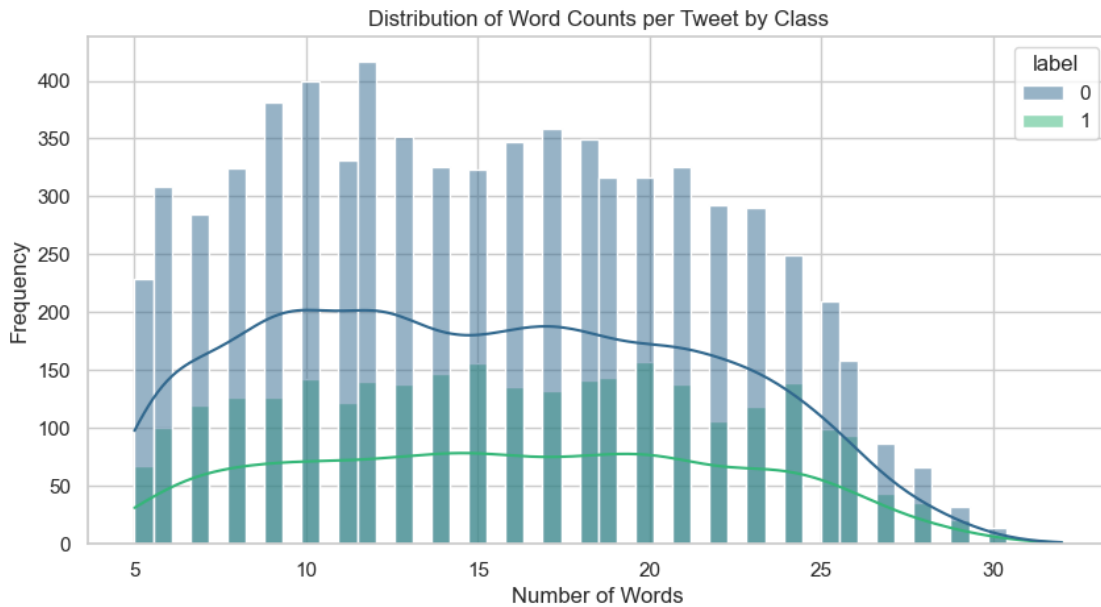
```
Class Counts:
label
0    70.973876
1    29.026124
Name: proportion, dtype: float64
```

1.2 1.2 Word Count Analysis

[8]:
```
df_train['word_count'] = df_train['tweet'].apply(lambda x: len(str(x).split()))

print("--- Overall Word Count Statistics ---")
print(df_train['word_count'].describe())
```

```
--- Overall Word Count Statistics ---
count    9991.000000
mean       15.567010
std         6.233004
min         5.000000
25%        10.000000
50%        15.000000
75%        21.000000
max        32.000000
Name: word_count, dtype: float64
```

[9]:
```
plt.figure(figsize=(10, 5))
sns.histplot(data=df_train, x='word_count', hue='label', bins=50, kde=True,␣
↪palette='viridis')
plt.title('Distribution of Word Counts per Tweet by Class')
plt.xlabel('Number of Words')
plt.ylabel('Frequency')
plt.show()
```



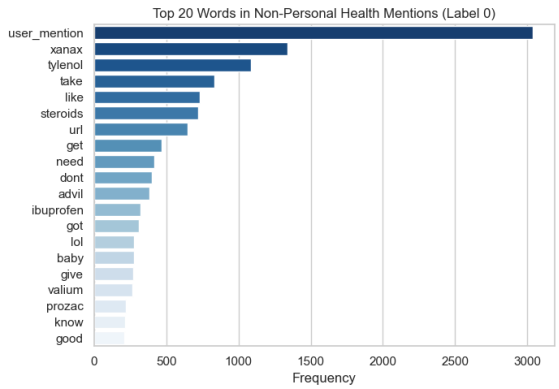
[10]:
```
df_class_0 = df_train[df_train['label'] == 0]
df_class_1 = df_train[df_train['label'] == 1]

print(f"\nAverage word count for Label 0: {df_class_0['word_count'].mean():.
↪2f}")
print(f"Average word count for Label 1: {df_class_1['word_count'].mean():.2f}")
```

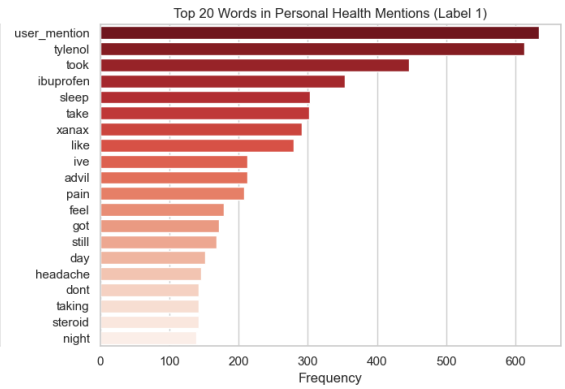
```
Average word count for Label 0: 15.35
Average word count for Label 1: 16.10
```

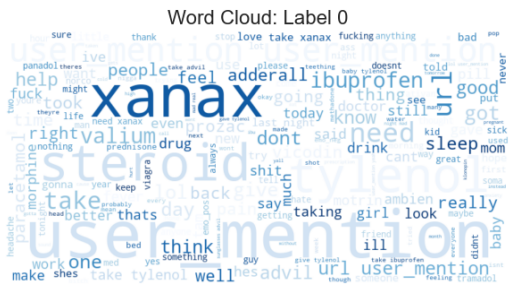
1.3 1.3 Vocabulary and Common Words Analysis

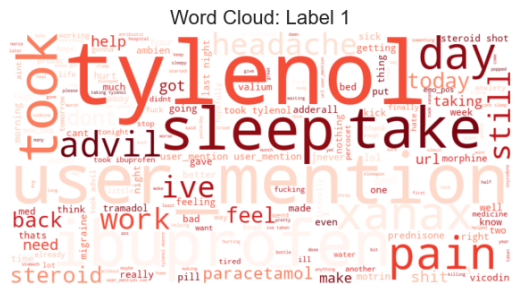
[11]:
```
def get_most_common_words(text_series, num_words=20):
    all_words = ' '.join(text_series.astype(str)).lower().split()
    filtered_words = [word for word in all_words if word not in stop_words and␣
↪len(word) > 2]
    word_freq = Counter(filtered_words)
    return word_freq.most_common(num_words), ' '.join(filtered_words)
```

```
common_0, text_0 = get_most_common_words(df_class_0['tweet'])
common_1, text_1 = get_most_common_words(df_class_1['tweet'])


x_0 = [val[1] for val in common_0]
y_0 = [val[0] for val in common_0]

x_1 = [val[1] for val in common_1]
y_1 = [val[0] for val in common_1]


plt.figure(figsize=(14, 5))
plt.subplot(1, 2, 1)

sns.barplot(x=x_0, y=y_0, hue=y_0, palette='Blues_r', legend=False)
plt.title('Top 20 Words in Non-Personal Health Mentions (Label 0)')
plt.xlabel('Frequency')


plt.subplot(1, 2, 2)
sns.barplot(x=x_1, y=y_1, hue=y_1, palette='Reds_r', legend=False)
plt.title('Top 20 Words in Personal Health Mentions (Label 1)')
plt.xlabel('Frequency')
plt.tight_layout()
plt.show()

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 8))

wc_0 = WordCloud(width=800, height=400, background_color='white',␣
 ↪colormap='Blues').generate(text_0)
ax1.imshow(wc_0, interpolation='bilinear')
ax1.set_title('Word Cloud: Label 0', fontsize=16)
ax1.axis('off')

wc_1 = WordCloud(width=800, height=400, background_color='white',␣
 ↪colormap='Reds').generate(text_1)
ax2.imshow(wc_1, interpolation='bilinear')
ax2.set_title('Word Cloud: Label 1', fontsize=16)
ax2.axis('off')

plt.show()
```









2 Phase 2: Text Preprocessing and PyTorch Dataset Setup

2.1 2.1 Python Code (Text Cleaning & Vocabulary Building)

[12]:
```
import re
import emoji
import contractions
from collections import Counter
from sklearn.model_selection import StratifiedKFold
from torch.utils.data import Dataset, DataLoader

def clean_tweet_advanced(text):
    text = str(text)
    text = contractions.fix(text)
    text = emoji.demojize(text)
    text = text.replace(':', ' ')
    text = text.lower()
    text = re.sub(r'[^a-z0-9_ ]', '', text)
    text = re.sub(r'\s+', ' ', text).strip()
    return text
```

```
df_train['clean_tweet'] = df_train['tweet'].apply(clean_tweet_advanced)

all_words = ' '.join(df_train['clean_tweet']).split()
word_counts = Counter(all_words)

min_freq = 2
vocab_words = [word for word, count in word_counts.items() if count >= min_freq]

word2idx = {'<PAD>': 0, '<UNK>': 1}
for idx, word in enumerate(vocab_words, start=2):
    word2idx[word] = idx

idx2word = {idx: word for word, idx in word2idx.items()}
VOCAB_SIZE = len(word2idx)

print(f"Original vocabulary size: {len(word_counts)}")
print(f"Reduced vocabulary size (min_freq={min_freq}): {VOCAB_SIZE}")

sample_messy_tweet = "I don't feel well today  took some advil  "
print(f"\nBefore: {sample_messy_tweet}")
print(f"After : {clean_tweet_advanced(sample_messy_tweet)}")
```

```
Original vocabulary size: 12821
Reduced vocabulary size (min_freq=2): 5359

Before: I don't feel well today  took some advil
After : i do not feel well today face_with_thermometer took some advil pill
```

2.2 2.2 Python Code (Tokenization and PyTorch Dataset)

[13]:
```
MAX_LEN = 40

def encode_tweet(text, vocab, max_len):
    tokens = text.split()
    encoded = [vocab.get(word, 1) for word in tokens]

    if len(encoded) < max_len:
        encoded = encoded + [0] * (max_len - len(encoded))
    else:
        encoded = encoded[:max_len]

    return encoded

class TweetDataset(Dataset):
    def __init__(self, texts, labels, vocab, max_len):
        self.texts = texts
        self.labels = labels
```

```
        self.vocab = vocab
        self.max_len = max_len

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        text = self.texts.iloc[idx]
        label = self.labels.iloc[idx]

        encoded_text = encode_tweet(text, self.vocab, self.max_len)

        x_tensor = torch.tensor(encoded_text, dtype=torch.long)
        y_tensor = torch.tensor(label, dtype=torch.float32)

        return x_tensor, y_tensor

sample_dataset = TweetDataset(df_train['clean_tweet'], df_train['label'],␣
 ↪word2idx, MAX_LEN)
sample_x, sample_y = sample_dataset[0]
print("--- Dataset Output Test ---")
print(f"Original Text: {df_train['clean_tweet'].iloc[0]}")
print(f"Encoded Tensor: {sample_x}")
print(f"Label Tensor: {sample_y}")
```

```
--- Dataset Output Test ---
Original Text: user_mention all i can tell you is i have had few relapses on it
it is not a cure i hear if you do not test for the jc virus tysabri is good
Encoded Tensor: tensor([ 2,  3,  4,  5,  6,  7,  8,  4,  9, 10, 11, 12, 13, 14,
14,  8, 15, 16,
        17,  4, 18, 19,  7, 20, 15, 21, 22, 23, 24, 25, 26,  8, 27,  0,  0,  0,
         0,  0,  0,  0])
Label Tensor: 0.0
```

2.3 2.3 Python Code (5-Fold Setup & Class Weights)

[14]:
```
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

X_data = df_train['clean_tweet']
y_data = df_train['label']

class_0_count = len(df_train[df_train['label'] == 0])
class_1_count = len(df_train[df_train['label'] == 1])
total_count = len(df_train)

weight_for_0 = (1 / class_0_count) * (total_count / 2.0)
weight_for_1 = (1 / class_1_count) * (total_count / 2.0)
```

```
pos_weight = torch.tensor([weight_for_1 / weight_for_0]).to(device)

print(f"Calculated Positive Class Weight (Multiplier for Label 1): {pos_weight.
 ↪item():.4f}")
print("5-Fold Cross Validation is ready for the training loop.")
```

```
Calculated Positive Class Weight (Multiplier for Label 1): 2.4452
5-Fold Cross Validation is ready for the training loop.
```

3 Phase 3: Building the LSTM and Bi-LSTM Models

[15]:
```
import torch
import torch.nn as nn

class AdvancedTweetClassifierLSTM(nn.Module):
    def __init__(self, vocab_size, embedding_dim, hidden_dim, output_dim,␣
 ↪n_layers,
                 bidirectional, embed_drop, lstm_drop, fc_drop, fc_hidden_dim):
        super(AdvancedTweetClassifierLSTM, self).__init__()


        self.embedding = nn.Embedding(vocab_size, embedding_dim, padding_idx=0)
        self.embed_dropout = nn.Dropout(embed_drop)


        self.lstm = nn.LSTM(embedding_dim,
                            hidden_dim,
                            num_layers=n_layers,
                            bidirectional=bidirectional,
                            dropout=lstm_drop if n_layers > 1 else 0,
                            batch_first=True)


        self.lstm_out_dim = hidden_dim * 2 if bidirectional else hidden_dim


        self.layer_norm = nn.LayerNorm(self.lstm_out_dim)


        self.fc_dropout = nn.Dropout(fc_drop)
        self.fc1 = nn.Linear(self.lstm_out_dim, fc_hidden_dim)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(fc_hidden_dim, output_dim)

    def forward(self, text):
```

```
        embedded = self.embed_dropout(self.embedding(text))

        output, (hidden, cell) = self.lstm(embedded)

        pooled_output, _ = torch.max(output, dim=1)

        pooled_output = self.layer_norm(pooled_output)
        pooled_output = self.fc_dropout(pooled_output)

        dense_out = self.fc1(pooled_output)
        dense_out = self.relu(dense_out)
        dense_out = self.fc_dropout(dense_out)

        return self.fc2(dense_out)

print("Advanced LSTM with Global Max Pooling defined successfully!")
```

```
Advanced LSTM with Global Max Pooling defined successfully!
```

4 Phase 4: Advanced Hyperparameter Tuning with Optuna

4.1 4.1 Python Code (Helper Functions & Metrics)

[16]:
```
import torch.optim as optim
from sklearn.metrics import f1_score, accuracy_score
from torch.cuda.amp import autocast, GradScaler
import optuna
import copy

class FocalLoss(nn.Module):
    def __init__(self, alpha=1, gamma=2, logits=True, reduce=True):
        super(FocalLoss, self).__init__()
        self.alpha = alpha
        self.gamma = gamma
        self.bce_with_logits = nn.BCEWithLogitsLoss(reduction='none')

    def forward(self, inputs, targets):
        BCE_loss = self.bce_with_logits(inputs, targets)
        pt = torch.exp(-BCE_loss)
        F_loss = self.alpha * (1-pt)**self.gamma * BCE_loss
        return torch.mean(F_loss)

print("Advanced Metrics and Focal Loss ready.")
```

```
Advanced Metrics and Focal Loss ready.
```

4.2 4.2 Python Code (The Optuna Objective Function)

[17]:
```
import torch.optim as optim
from torch.optim.lr_scheduler import CosineAnnealingLR
from sklearn.metrics import f1_score, accuracy_score
from torch.cuda.amp import autocast, GradScaler
import optuna
import numpy as np

MAX_EPOCHS = 10

def objective(trial):
    embed_dim = trial.suggest_categorical('embedding_dim', [100, 200, 300])
    hidden_dim = trial.suggest_categorical('hidden_dim', [64, 128, 256])
    fc_hidden_dim = trial.suggest_categorical('fc_hidden_dim', [32, 64, 128])
    lr = trial.suggest_float('lr', 1e-4, 5e-3, log=True)
    weight_decay = trial.suggest_float('weight_decay', 1e-5, 1e-2, log=True)
    embed_drop = trial.suggest_float('embed_drop', 0.2, 0.6)
    lstm_drop = trial.suggest_float('lstm_drop', 0.2, 0.6)
    fc_drop = trial.suggest_float('fc_drop', 0.2, 0.6)
    batch_size = trial.suggest_categorical('batch_size', [32, 64, 128])
    is_bidirectional = trial.suggest_categorical('bidirectional', [True, False])

    gamma = trial.suggest_float('gamma', 1.0, 3.0)

    fold_f1_scores = []

    for fold, (train_idx, val_idx) in enumerate(skf.split(X_data, y_data)):

        train_texts, val_texts = X_data.iloc[train_idx], X_data.iloc[val_idx]
        train_labels, val_labels = y_data.iloc[train_idx], y_data.iloc[val_idx]

        train_dataset = TweetDataset(train_texts, train_labels, word2idx,␣
↪MAX_LEN)
        val_dataset = TweetDataset(val_texts, val_labels, word2idx, MAX_LEN)

        train_loader = DataLoader(train_dataset, batch_size=batch_size,␣
↪shuffle=True)
        val_loader = DataLoader(val_dataset, batch_size=batch_size,␣
↪shuffle=False)

        model = AdvancedTweetClassifierLSTM(
            vocab_size=VOCAB_SIZE,
            embedding_dim=embed_dim,
            hidden_dim=hidden_dim,
            output_dim=1,
            n_layers=2,
```

```
            bidirectional=is_bidirectional,
            embed_drop=embed_drop,
            lstm_drop=lstm_drop,
            fc_drop=fc_drop,
            fc_hidden_dim=fc_hidden_dim
        ).to(device)

        optimizer = optim.AdamW(model.parameters(), lr=lr,␣
↪weight_decay=weight_decay)

        scheduler = CosineAnnealingLR(optimizer, T_max=MAX_EPOCHS)

        criterion = FocalLoss(alpha=pos_weight.item(), gamma=gamma)
        scaler = GradScaler()

        best_val_f1 = 0
        patience_counter = 0
        PATIENCE = 3

        for epoch in range(MAX_EPOCHS):
            model.train()
            for texts, labels in train_loader:
                texts, labels = texts.to(device), labels.to(device)
                optimizer.zero_grad()
                with autocast():
                    predictions = model(texts).squeeze(1)
                    loss = criterion(predictions, labels)
                scaler.scale(loss).backward()
                scaler.step(optimizer)
                scaler.update()

            scheduler.step()

            model.eval()
            all_preds, all_labels = [], []
            with torch.no_grad():
                for texts, labels in val_loader:
                    texts, labels = texts.to(device), labels.to(device)
                    with autocast():
                        predictions = model(texts).squeeze(1)
                    probs = torch.sigmoid(predictions)
                    preds = (probs > 0.5).float()
                    all_preds.extend(preds.cpu().numpy())
                    all_labels.extend(labels.cpu().numpy())

            epoch_f1 = f1_score(all_labels, all_preds, average='macro')
```

```
            if epoch_f1 > best_val_f1:
                best_val_f1 = epoch_f1
                patience_counter = 0
            else:
                patience_counter += 1

            if patience_counter >= PATIENCE:
                break

        fold_f1_scores.append(best_val_f1)

    return np.mean(fold_f1_scores)

print("Upgraded Optuna Objective Function ready!")
```

```
Upgraded Optuna Objective Function ready!
```

4.3 4.3 Python Code (Running the Optimization)

[18]:
```
import warnings
warnings.filterwarnings('ignore')

print("Starting Advanced Optuna Search with Early Stopping & AdamW...")

study = optuna.create_study(direction='maximize',␣
 ↪study_name="Tweet_LSTM_Advanced_Tuning")

study.optimize(objective, n_trials=25)

print("\n=== ADVANCED OPTIMIZATION FINISHED ===")
print(f"Best Macro F1-Score: {study.best_value:.4f}")
print("Best Hyperparameters:")
for key, value in study.best_params.items():
    print(f"    {key}: {value}")
```

```
[I 2026-03-01 20:20:04,073] A new study created in memory with name:
Tweet_LSTM_Advanced_Tuning

Starting Advanced Optuna Search with Early Stopping & AdamW…

[I 2026-03-01 20:23:07,319] Trial 0 finished with value:
0.7349229039184968 and parameters: {'embedding_dim': 100, 'hidden_dim': 256,
'fc_hidden_dim': 64, 'lr': 0.00010931223797726832, 'weight_decay':
0.0003585571215430628, 'embed_drop': 0.5224541451506037, 'lstm_drop':
0.25500941268092764, 'fc_drop': 0.32769139033350425, 'batch_size': 64,
'bidirectional': True, 'gamma': 2.1385381262649386}. Best is trial 0 with value:
0.7349229039184968.
[I 2026-03-01 20:27:03,189] Trial 1 finished with value:
0.7979072301830955 and parameters: {'embedding_dim': 100, 'hidden_dim': 256,
```

```
'fc_hidden_dim': 32, 'lr': 0.00033879962461157034, 'weight_decay':
7.45442342631186e-05, 'embed_drop': 0.38411304776156796, 'lstm_drop':
0.24501825796254573, 'fc_drop': 0.3037006262383604, 'batch_size': 32,
'bidirectional': False, 'gamma': 1.6446532231389692}. Best is trial 1 with
value: 0.7979072301830955.
[I 2026-03-01 20:27:50,043] Trial 2 finished with value:
0.722691144151195 and parameters: {'embedding_dim': 200, 'hidden_dim': 64,
'fc_hidden_dim': 128, 'lr': 0.0003360751129713213, 'weight_decay':
0.0013568910759299072, 'embed_drop': 0.5471630802953109, 'lstm_drop':
0.2921565948998184, 'fc_drop': 0.4341250225156782, 'batch_size': 128,
'bidirectional': False, 'gamma': 1.3481105468061256}. Best is trial 1 with
value: 0.7979072301830955.
[I 2026-03-01 20:30:11,414] Trial 3 finished with value:
0.8093416485561568 and parameters: {'embedding_dim': 300, 'hidden_dim': 64,
'fc_hidden_dim': 32, 'lr': 0.0016699776040712455, 'weight_decay':
1.3220214556716614e-05, 'embed_drop': 0.4193563756146569, 'lstm_drop':
0.39802724757164476, 'fc_drop': 0.2888397796167249, 'batch_size': 32,
'bidirectional': True, 'gamma': 1.3811279927232067}. Best is trial 3 with value:
0.8093416485561568.
[I 2026-03-01 20:31:46,637] Trial 4 finished with value:
0.715924737791903 and parameters: {'embedding_dim': 100, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.00020302247190485595, 'weight_decay':
0.005594879379707951, 'embed_drop': 0.5562343700525145, 'lstm_drop':
0.3244113038806125, 'fc_drop': 0.4809033941640666, 'batch_size': 128,
'bidirectional': True, 'gamma': 1.1487804828508177}. Best is trial 3 with value:
0.8093416485561568.
[I 2026-03-01 20:32:44,150] Trial 5 finished with value:
0.8100680240482392 and parameters: {'embedding_dim': 300, 'hidden_dim': 64,
'fc_hidden_dim': 32, 'lr': 0.002294447593872326, 'weight_decay':
0.0016100387828159206, 'embed_drop': 0.23435125783381575, 'lstm_drop':
0.25873098771606085, 'fc_drop': 0.323417231787336, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.785515599379818}. Best is trial 5 with value:
0.8100680240482392.
[I 2026-03-01 20:34:01,342] Trial 6 finished with value:
0.8053994368795931 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 64, 'lr': 0.0014572862865885787, 'weight_decay':
0.006670118645373446, 'embed_drop': 0.27436613943885835, 'lstm_drop':
0.5944849131072962, 'fc_drop': 0.4875880864060941, 'batch_size': 64,
'bidirectional': False, 'gamma': 2.644305460577622}. Best is trial 5 with value:
0.8100680240482392.
[I 2026-03-01 20:35:44,856] Trial 7 finished with value:
0.8138027191917618 and parameters: {'embedding_dim': 300, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.002198849241993338, 'weight_decay':
0.00011249910467606451, 'embed_drop': 0.5890812041249789, 'lstm_drop':
0.23001651066446344, 'fc_drop': 0.20149600931342257, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.165391266321242}. Best is trial 7 with value:
0.8138027191917618.
[I 2026-03-01 20:36:38,900] Trial 8 finished with value:
```

```
0.7914136796532768 and parameters: {'embedding_dim': 100, 'hidden_dim': 64,
'fc_hidden_dim': 128, 'lr': 0.0013924028055317653, 'weight_decay':
1.2739094635880739e-05, 'embed_drop': 0.24094188697329103, 'lstm_drop':
0.3023974630866643, 'fc_drop': 0.5607964794295395, 'batch_size': 128,
'bidirectional': False, 'gamma': 1.5767766410664343}. Best is trial 7 with
value: 0.8138027191917618.
[I 2026-03-01 20:42:39,796] Trial 9 finished with value:
0.8090105493321676 and parameters: {'embedding_dim': 200, 'hidden_dim': 256,
'fc_hidden_dim': 128, 'lr': 0.0016693472539057997, 'weight_decay':
0.0008794649566646196, 'embed_drop': 0.3728730059313774, 'lstm_drop':
0.28495137893502276, 'fc_drop': 0.3280452851233385, 'batch_size': 32,
'bidirectional': True, 'gamma': 1.9412301291340284}. Best is trial 7 with value:
0.8138027191917618.
[I 2026-03-01 20:43:29,629] Trial 10 finished with value:
0.8140772809437866 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 32, 'lr': 0.004435776757597687, 'weight_decay':
5.3426978971911836e-05, 'embed_drop': 0.47060712833815865, 'lstm_drop':
0.5376721267174642, 'fc_drop': 0.21594546570766224, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.2458778184031134}. Best is trial 10 with
value: 0.8140772809437866.
[I 2026-03-01 20:44:21,165] Trial 11 finished with value:
0.8114394519588837 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 32, 'lr': 0.0038929161053375986, 'weight_decay':
7.52480378289649e-05, 'embed_drop': 0.4727796304051732, 'lstm_drop':
0.5576589064490312, 'fc_drop': 0.2035437696304801, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.3100126580506863}. Best is trial 10 with
value: 0.8140772809437866.
[I 2026-03-01 20:45:11,711] Trial 12 finished with value:
0.8082008939952725 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 32, 'lr': 0.004298686844113047, 'weight_decay':
5.130770963133543e-05, 'embed_drop': 0.593611728799909, 'lstm_drop':
0.5075670333916843, 'fc_drop': 0.20244504999949675, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.4304326432270797}. Best is trial 10 with
value: 0.8140772809437866.
[I 2026-03-01 20:46:09,426] Trial 13 finished with value:
0.8010844720467573 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 32, 'lr': 0.0007003678085545287, 'weight_decay':
0.00017197445924253894, 'embed_drop': 0.47172580073547943, 'lstm_drop':
0.45810454977851905, 'fc_drop': 0.23432162598765205, 'batch_size': 128,
'bidirectional': True, 'gamma': 1.8380902620451685}. Best is trial 10 with
value: 0.8140772809437866.
[I 2026-03-01 20:47:53,702] Trial 14 finished with value:
0.8188067193619638 and parameters: {'embedding_dim': 300, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.0031211170983288052, 'weight_decay':
3.4013355922825807e-05, 'embed_drop': 0.5984435787610937, 'lstm_drop':
0.3895735526295105, 'fc_drop': 0.26012005216339595, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.4601944153295756}. Best is trial 14 with
value: 0.8188067193619638.
```

```
[I 2026-03-01 20:49:21,608] Trial 15 finished with value:
0.8124877049567205 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 64, 'lr': 0.0044542839974914955, 'weight_decay':
2.9493745663691204e-05, 'embed_drop': 0.48499347679422317, 'lstm_drop':
0.3813399935976087, 'fc_drop': 0.26492760790761066, 'batch_size': 64,
'bidirectional': True, 'gamma': 2.997433310371204}. Best is trial 14 with value:
0.8188067193619638.
[I 2026-03-01 20:51:04,409] Trial 16 finished with value:
0.8046021307975844 and parameters: {'embedding_dim': 200, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.000720670367973917, 'weight_decay':
2.7156522263422683e-05, 'embed_drop': 0.333407045251488, 'lstm_drop':
0.4544051316904718, 'fc_drop': 0.4205028836449454, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.5301910723399437}. Best is trial 14 with
value: 0.8188067193619638.
[I 2026-03-01 20:52:07,084] Trial 17 finished with value:
0.8136655655381443 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 32, 'lr': 0.0028432875865603124, 'weight_decay':
0.0002870494553448806, 'embed_drop': 0.4292014714800444, 'lstm_drop':
0.47397266893369755, 'fc_drop': 0.3620815991782119, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.1399044659998014}. Best is trial 14 with
value: 0.8188067193619638.
[I 2026-03-01 20:53:24,549] Trial 18 finished with value:
0.8105137631838464 and parameters: {'embedding_dim': 300, 'hidden_dim': 128,
'fc_hidden_dim': 128, 'lr': 0.0008810889322632249, 'weight_decay':
2.7739545449755885e-05, 'embed_drop': 0.31915465142514865, 'lstm_drop':
0.34408744357212756, 'fc_drop': 0.2605938199807478, 'batch_size': 64,
'bidirectional': False, 'gamma': 2.8263643787940507}. Best is trial 14 with
value: 0.8188067193619638.
[I 2026-03-01 20:59:46,013] Trial 19 finished with value:
0.8131307601165532 and parameters: {'embedding_dim': 200, 'hidden_dim': 256,
'fc_hidden_dim': 64, 'lr': 0.003301815299517484, 'weight_decay':
0.00020906041420154346, 'embed_drop': 0.5152426804142471, 'lstm_drop':
0.5218470551055449, 'fc_drop': 0.372890324113031, 'batch_size': 32,
'bidirectional': True, 'gamma': 2.351242364114056}. Best is trial 14 with value:
0.8188067193619638.
[I 2026-03-01 21:01:23,621] Trial 20 finished with value:
0.8156987203197504 and parameters: {'embedding_dim': 300, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.0010533962898880657, 'weight_decay':
0.0004796219515509771, 'embed_drop': 0.43664868611904994, 'lstm_drop':
0.4173058642320054, 'fc_drop': 0.25244023295093554, 'batch_size': 128,
'bidirectional': True, 'gamma': 1.78607931567683}. Best is trial 14 with value:
0.8188067193619638.
[I 2026-03-01 21:03:11,228] Trial 21 finished with value:
0.8136407606062301 and parameters: {'embedding_dim': 300, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.0009696627377977918, 'weight_decay':
0.0007529767072502583, 'embed_drop': 0.44644600877693824, 'lstm_drop':
0.42412163801656033, 'fc_drop': 0.24947233741210006, 'batch_size': 128,
'bidirectional': True, 'gamma': 1.8258641078319542}. Best is trial 14 with
```

```
value: 0.8188067193619638.
[I 2026-03-01 21:04:45,828] Trial 22 finished with value:
0.8076193505725258 and parameters: {'embedding_dim': 300, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.0025538015755767578, 'weight_decay':
0.0006401541651889669, 'embed_drop': 0.3381815030762936, 'lstm_drop':
0.3581983865941615, 'fc_drop': 0.23928701190384968, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.0039144083433906}. Best is trial 14 with
value: 0.8188067193619638.
[I 2026-03-01 21:06:44,769] Trial 23 finished with value:
0.8026570797551436 and parameters: {'embedding_dim': 300, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.00041748659374809925, 'weight_decay':
4.789566382790653e-05, 'embed_drop': 0.3956327419351966, 'lstm_drop':
0.4270724354171621, 'fc_drop': 0.28172663056015035, 'batch_size': 128,
'bidirectional': True, 'gamma': 1.631508438282899}. Best is trial 14 with value:
0.8188067193619638.
[I 2026-03-01 21:08:36,869] Trial 24 finished with value:
0.8097791020252499 and parameters: {'embedding_dim': 300, 'hidden_dim': 256,
'fc_hidden_dim': 32, 'lr': 0.0011904651926272784, 'weight_decay':
0.0027287837291330493, 'embed_drop': 0.495085589481807, 'lstm_drop':
0.4967284872626491, 'fc_drop': 0.3681628659845049, 'batch_size': 128,
'bidirectional': True, 'gamma': 2.5816797313513526}. Best is trial 14 with
value: 0.8188067193619638.


=== ADVANCED OPTIMIZATION FINISHED ===
Best Macro F1-Score: 0.8188
Best Hyperparameters:
    embedding_dim: 300
    hidden_dim: 256
    fc_hidden_dim: 32
    lr: 0.0031211170983288052
    weight_decay: 3.4013355922825807e-05
    embed_drop: 0.5984435787610937
    lstm_drop: 0.3895735526295105
    fc_drop: 0.26012005216339595
    batch_size: 128
    bidirectional: True
    gamma: 2.4601944153295756
```

5 Phase 5: Final Model Training, Comparison, and Saving

[26]:
```
import pandas as pd
import torch
import torch.optim as optim
from torch.cuda.amp import autocast, GradScaler
from sklearn.metrics import accuracy_score, classification_report,␣
 ↪confusion_matrix
```

```python
import matplotlib.pyplot as plt
import seaborn as sns
import copy

BEST_PARAMS = study.best_params

print("--- Preparing 100% Train and Test Data ---")
df_test = pd.read_csv('phm_test.csv')
df_test = df_test.dropna(subset=['tweet']).reset_index(drop=True)

df_test['clean_tweet'] = df_test['tweet'].apply(clean_tweet_advanced)

train_dataset_final = TweetDataset(df_train['clean_tweet'], df_train['label'],␣
↪word2idx, MAX_LEN)
test_dataset_final = TweetDataset(df_test['clean_tweet'], df_test['label'],␣
↪word2idx, MAX_LEN)

train_loader_final = DataLoader(train_dataset_final,␣
↪batch_size=BEST_PARAMS['batch_size'], shuffle=True)
test_loader_final = DataLoader(test_dataset_final,␣
↪batch_size=BEST_PARAMS['batch_size'], shuffle=False)

FINAL_EPOCHS = 30

def train_and_evaluate_model(is_bidirectional, model_name, save_filename):
    print(f"\n=========================================================")
    print(f"   Training & Testing Model: {model_name}")
    print(f"=========================================================")

    model = AdvancedTweetClassifierLSTM(
        vocab_size=VOCAB_SIZE,
        embedding_dim=BEST_PARAMS['embedding_dim'],
        hidden_dim=BEST_PARAMS['hidden_dim'],
        output_dim=1,
        n_layers=2,
        bidirectional=is_bidirectional,
        embed_drop=BEST_PARAMS['embed_drop'],
        lstm_drop=BEST_PARAMS['lstm_drop'],
        fc_drop=BEST_PARAMS['fc_drop'],
        fc_hidden_dim=BEST_PARAMS['fc_hidden_dim']
    ).to(device)

    # For final training, we reduce the learning rate to 15% for better␣
↪convergence
    optimizer = optim.AdamW(model.parameters(), lr=BEST_PARAMS['lr'] *0.15,␣
↪weight_decay=BEST_PARAMS['weight_decay'])
```

```
    scheduler = optim.lr_scheduler.CosineAnnealingLR(optimizer,␣
↪T_max=FINAL_EPOCHS)
    criterion = FocalLoss(alpha=pos_weight.item(), gamma=BEST_PARAMS['gamma'])
    scaler = GradScaler()

    history = {'train_loss': [], 'test_loss': [], 'test_acc': [], 'train_acc':␣
↪[]}

    best_test_acc = 0.0
    best_model_weights = None
    best_epoch = 0
    best_test_preds = []
    best_test_labels = []

    for epoch in range(FINAL_EPOCHS):
        model.train()
        epoch_train_loss = 0
        all_train_preds, all_train_labels = [], []

        for texts, labels in train_loader_final:
            texts, labels = texts.to(device), labels.to(device)
            optimizer.zero_grad()
            with autocast():
                predictions = model(texts).squeeze(1)
                loss = criterion(predictions, labels)

            scaler.scale(loss).backward()
            scaler.step(optimizer)
            scaler.update()
            epoch_train_loss += loss.item()

            probs = torch.sigmoid(predictions)
            preds = (probs > 0.5).float()
            all_train_preds.extend(preds.detach().cpu().numpy())
            all_train_labels.extend(labels.cpu().numpy())

        scheduler.step()
        avg_train_loss = epoch_train_loss / len(train_loader_final)
        train_acc = accuracy_score(all_train_labels, all_train_preds)
        history['train_loss'].append(avg_train_loss)
        history['train_acc'].append(train_acc)

        model.eval()
        epoch_test_loss = 0
        all_test_preds, all_test_labels = [], []

        with torch.no_grad():
```

```
            for texts, labels in test_loader_final:
                texts, labels = texts.to(device), labels.to(device)
                with autocast():
                    predictions = model(texts).squeeze(1)
                    test_loss = criterion(predictions, labels)

                epoch_test_loss += test_loss.item()
                probs = torch.sigmoid(predictions)
                preds = (probs > 0.5).float()

                all_test_preds.extend(preds.cpu().numpy())
                all_test_labels.extend(labels.cpu().numpy())

        avg_test_loss = epoch_test_loss / len(test_loader_final)
        test_acc = accuracy_score(all_test_labels, all_test_preds)

        history['test_loss'].append(avg_test_loss)
        history['test_acc'].append(test_acc)

        if test_acc > best_test_acc:
            best_test_acc = test_acc
            best_epoch = epoch + 1
            best_model_weights = copy.deepcopy(model.state_dict())
            best_test_preds = all_test_preds
            best_test_labels = all_test_labels
            save_msg = "  <-- [NEW BEST TEST ACCURACY SAVED]"
        else:
            save_msg = ""

        print(f"Epoch {epoch+1:02d}/{FINAL_EPOCHS} | Train Loss:␣
↪{avg_train_loss:.4f} | Test Loss: {avg_test_loss:.4f} | Test Acc: {test_acc:.
↪4f}{save_msg}")

    torch.save(best_model_weights, save_filename)
    print(f"\n>>> Finished! Saved {model_name} from Epoch {best_epoch} (Acc:␣
↪{best_test_acc:.4f}) as '{save_filename}'")

    print(f"\nFinal Classification Report for {model_name}:")
    print(classification_report(best_test_labels, best_test_preds,␣
↪target_names=['Non-Personal (0)', 'Personal (1)']))

    return history, best_test_preds, best_test_labels

history_lstm, lstm_preds, test_labels = train_and_evaluate_model(False,␣
↪"Standard LSTM", "LSTM_final_model.pth")
history_bilstm, bilstm_preds, _ = train_and_evaluate_model(True, "Bi-LSTM",␣
↪"BiLSTM_final_model.pth")
```

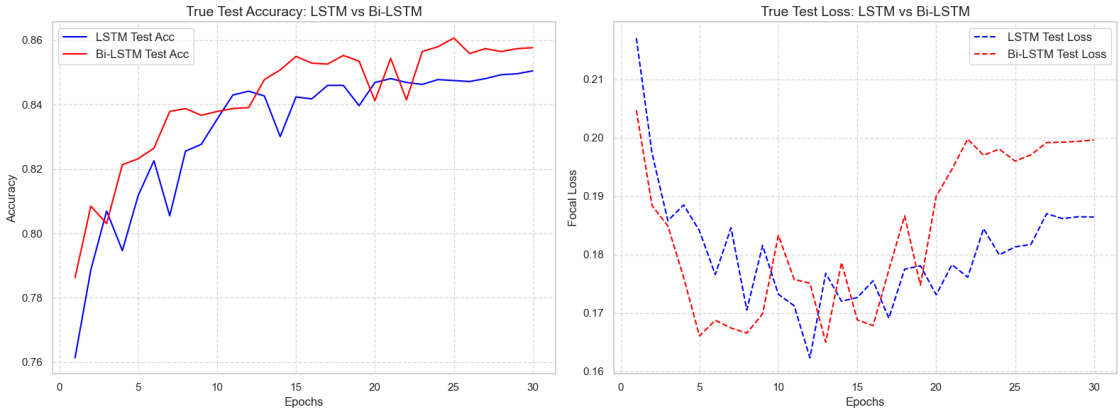
```
epochs_range = range(1, FINAL_EPOCHS + 1)
plt.figure(figsize=(16, 6))

plt.subplot(1, 2, 1)
plt.plot(epochs_range, history_lstm['test_acc'], label='LSTM Test Acc',␣
 ↪color='blue')
plt.plot(epochs_range, history_bilstm['test_acc'], label='Bi-LSTM Test Acc',␣
 ↪color='red')
plt.title('True Test Accuracy: LSTM vs Bi-LSTM', fontsize=14)
plt.xlabel('Epochs', fontsize=12)
plt.ylabel('Accuracy', fontsize=12)
plt.legend(fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7)

plt.subplot(1, 2, 2)
plt.plot(epochs_range, history_lstm['test_loss'], label='LSTM Test Loss',␣
 ↪color='blue', linestyle='--')
plt.plot(epochs_range, history_bilstm['test_loss'], label='Bi-LSTM Test Loss',␣
 ↪color='red', linestyle='--')
plt.title('True Test Loss: LSTM vs Bi-LSTM', fontsize=14)
plt.xlabel('Epochs', fontsize=12)
plt.ylabel('Focal Loss', fontsize=12)
plt.legend(fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7)

plt.tight_layout()
plt.savefig('final_model_comparison.png', dpi=300)
plt.show()

def plot_cm(labels, preds, name, filename, cmap):
    cm = confusion_matrix(labels, preds)
    plt.figure(figsize=(6, 5))
    sns.heatmap(cm, annot=True, fmt='d', cmap=cmap, xticklabels=['Non-Personal␣
 ↪(0)', 'Personal (1)'], yticklabels=['Non-Personal (0)', 'Personal (1)'])
    plt.title(f'Test Set Confusion Matrix ({name})')
    plt.ylabel('Actual Label')
    plt.xlabel('Predicted Label')
    plt.tight_layout()
    plt.savefig(filename, dpi=300)
    plt.show()

plot_cm(test_labels, lstm_preds, "Standard LSTM", "confusion_matrix_LSTM.png",␣
 ↪'Purples')
plot_cm(test_labels, bilstm_preds, "Bi-LSTM", "confusion_matrix_BiLSTM.png",␣
 ↪'Blues')
```

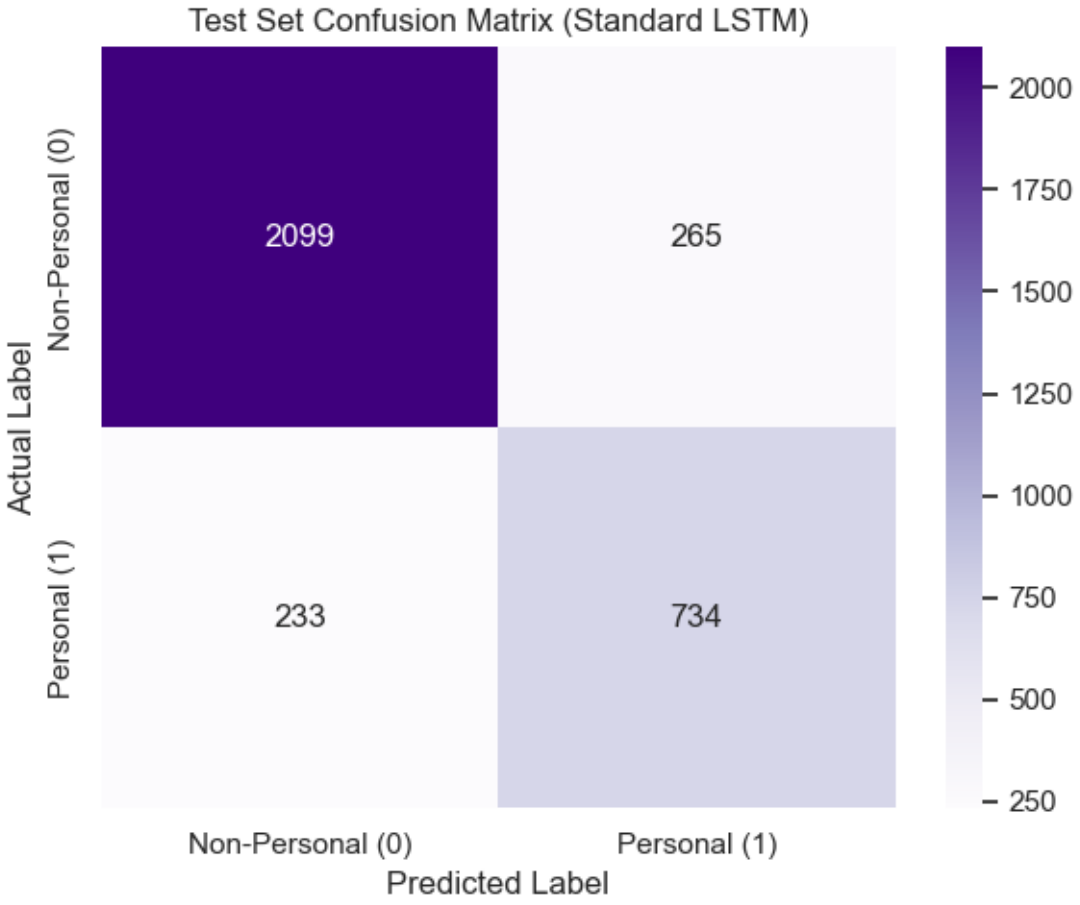
```python
df_test['LSTM_prediction'] = [int(p) for p in lstm_preds]
df_test['BiLSTM_prediction'] = [int(p) for p in bilstm_preds]
df_test.to_csv('phm_test_predictions_final.csv', index=False)
print("\n>>> All plots and predictions successfully saved!")
```

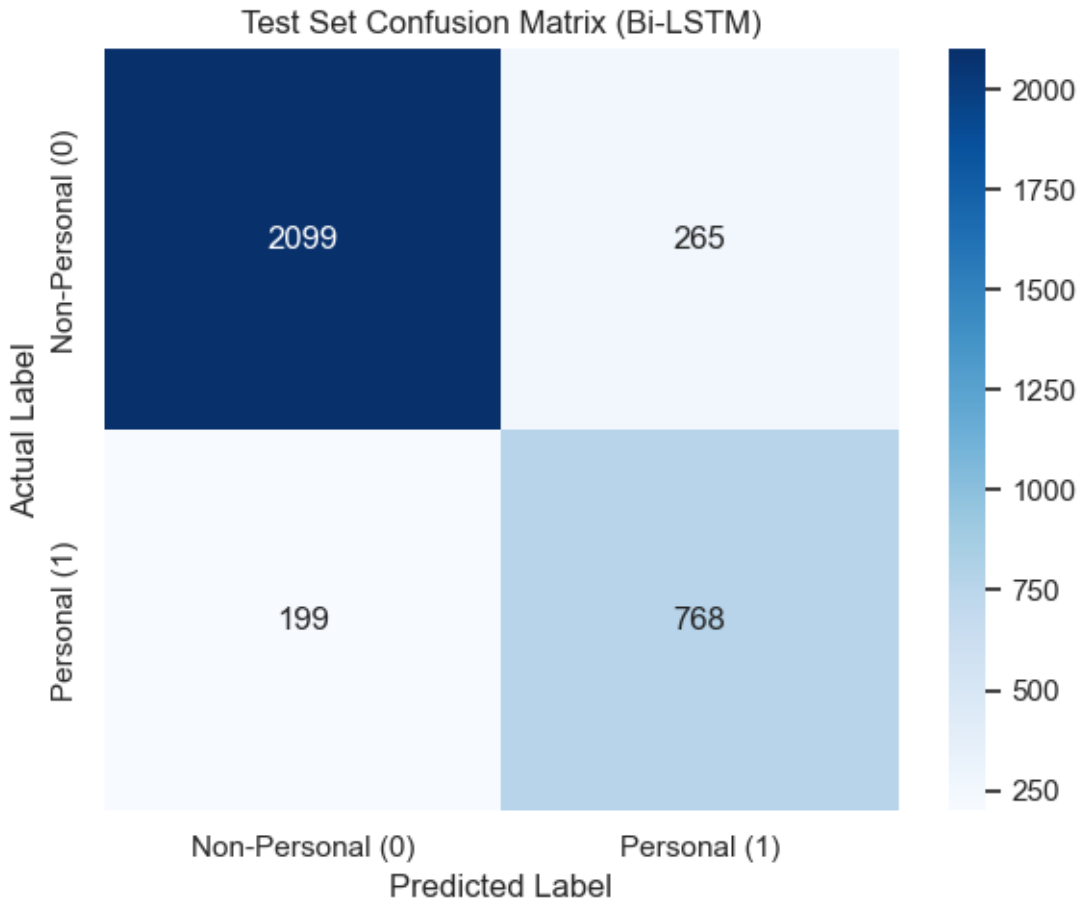
```
--- Preparing 100% Train and Test Data ---

=======================================================
   Training & Testing Model: Standard LSTM
=======================================================
Epoch 01/30 | Train Loss: 0.2565 | Test Loss: 0.2171 | Test Acc: 0.7613  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 02/30 | Train Loss: 0.2292 | Test Loss: 0.1973 | Test Acc: 0.7887  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 03/30 | Train Loss: 0.2136 | Test Loss: 0.1858 | Test Acc: 0.8070  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 04/30 | Train Loss: 0.2057 | Test Loss: 0.1885 | Test Acc: 0.7947
Epoch 05/30 | Train Loss: 0.1971 | Test Loss: 0.1841 | Test Acc: 0.8118  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 06/30 | Train Loss: 0.1949 | Test Loss: 0.1766 | Test Acc: 0.8226  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 07/30 | Train Loss: 0.1858 | Test Loss: 0.1846 | Test Acc: 0.8055
Epoch 08/30 | Train Loss: 0.1815 | Test Loss: 0.1705 | Test Acc: 0.8256  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 09/30 | Train Loss: 0.1768 | Test Loss: 0.1816 | Test Acc: 0.8277  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 10/30 | Train Loss: 0.1726 | Test Loss: 0.1732 | Test Acc: 0.8355  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 11/30 | Train Loss: 0.1674 | Test Loss: 0.1712 | Test Acc: 0.8430  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 12/30 | Train Loss: 0.1638 | Test Loss: 0.1623 | Test Acc: 0.8442  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 13/30 | Train Loss: 0.1583 | Test Loss: 0.1768 | Test Acc: 0.8427
Epoch 14/30 | Train Loss: 0.1561 | Test Loss: 0.1720 | Test Acc: 0.8301
Epoch 15/30 | Train Loss: 0.1519 | Test Loss: 0.1727 | Test Acc: 0.8424
Epoch 16/30 | Train Loss: 0.1508 | Test Loss: 0.1755 | Test Acc: 0.8418
Epoch 17/30 | Train Loss: 0.1463 | Test Loss: 0.1691 | Test Acc: 0.8460  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 18/30 | Train Loss: 0.1452 | Test Loss: 0.1775 | Test Acc: 0.8460
Epoch 19/30 | Train Loss: 0.1442 | Test Loss: 0.1781 | Test Acc: 0.8397
Epoch 20/30 | Train Loss: 0.1387 | Test Loss: 0.1731 | Test Acc: 0.8469  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 21/30 | Train Loss: 0.1339 | Test Loss: 0.1783 | Test Acc: 0.8481  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 22/30 | Train Loss: 0.1350 | Test Loss: 0.1761 | Test Acc: 0.8469
Epoch 23/30 | Train Loss: 0.1340 | Test Loss: 0.1844 | Test Acc: 0.8463
Epoch 24/30 | Train Loss: 0.1349 | Test Loss: 0.1800 | Test Acc: 0.8478
```

```
Epoch 25/30 | Train Loss: 0.1291 | Test Loss: 0.1813 | Test Acc: 0.8475
Epoch 26/30 | Train Loss: 0.1310 | Test Loss: 0.1817 | Test Acc: 0.8472
Epoch 27/30 | Train Loss: 0.1259 | Test Loss: 0.1870 | Test Acc: 0.8481
Epoch 28/30 | Train Loss: 0.1252 | Test Loss: 0.1862 | Test Acc: 0.8493  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 29/30 | Train Loss: 0.1242 | Test Loss: 0.1865 | Test Acc: 0.8496  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 30/30 | Train Loss: 0.1277 | Test Loss: 0.1864 | Test Acc: 0.8505  <--
[NEW BEST TEST ACCURACY SAVED]

>>> Finished! Saved Standard LSTM from Epoch 30 (Acc: 0.8505) as
'LSTM_final_model.pth'

Final Classification Report for Standard LSTM:
                  precision    recall  f1-score   support

Non-Personal (0)       0.90      0.89      0.89      2364
    Personal (1)       0.73      0.76      0.75       967

        accuracy                           0.85      3331
       macro avg       0.82      0.82      0.82      3331
    weighted avg       0.85      0.85      0.85      3331


=======================================================
   Training & Testing Model: Bi-LSTM
=======================================================
Epoch 01/30 | Train Loss: 0.2552 | Test Loss: 0.2047 | Test Acc: 0.7863  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 02/30 | Train Loss: 0.2218 | Test Loss: 0.1884 | Test Acc: 0.8085  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 03/30 | Train Loss: 0.2109 | Test Loss: 0.1849 | Test Acc: 0.8031
Epoch 04/30 | Train Loss: 0.2021 | Test Loss: 0.1761 | Test Acc: 0.8214  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 05/30 | Train Loss: 0.1929 | Test Loss: 0.1661 | Test Acc: 0.8232  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 06/30 | Train Loss: 0.1832 | Test Loss: 0.1688 | Test Acc: 0.8265  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 07/30 | Train Loss: 0.1800 | Test Loss: 0.1674 | Test Acc: 0.8379  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 08/30 | Train Loss: 0.1728 | Test Loss: 0.1666 | Test Acc: 0.8388  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 09/30 | Train Loss: 0.1696 | Test Loss: 0.1699 | Test Acc: 0.8367
Epoch 10/30 | Train Loss: 0.1660 | Test Loss: 0.1833 | Test Acc: 0.8379
Epoch 11/30 | Train Loss: 0.1574 | Test Loss: 0.1758 | Test Acc: 0.8388
Epoch 12/30 | Train Loss: 0.1523 | Test Loss: 0.1751 | Test Acc: 0.8391  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 13/30 | Train Loss: 0.1482 | Test Loss: 0.1650 | Test Acc: 0.8478  <--
```

```
[NEW BEST TEST ACCURACY SAVED]
Epoch 14/30 | Train Loss: 0.1435 | Test Loss: 0.1787 | Test Acc: 0.8508  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 15/30 | Train Loss: 0.1397 | Test Loss: 0.1688 | Test Acc: 0.8550  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 16/30 | Train Loss: 0.1369 | Test Loss: 0.1679 | Test Acc: 0.8529
Epoch 17/30 | Train Loss: 0.1321 | Test Loss: 0.1774 | Test Acc: 0.8526
Epoch 18/30 | Train Loss: 0.1277 | Test Loss: 0.1867 | Test Acc: 0.8553  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 19/30 | Train Loss: 0.1291 | Test Loss: 0.1748 | Test Acc: 0.8535
Epoch 20/30 | Train Loss: 0.1277 | Test Loss: 0.1900 | Test Acc: 0.8412
Epoch 21/30 | Train Loss: 0.1194 | Test Loss: 0.1946 | Test Acc: 0.8544
Epoch 22/30 | Train Loss: 0.1178 | Test Loss: 0.1998 | Test Acc: 0.8415
Epoch 23/30 | Train Loss: 0.1184 | Test Loss: 0.1970 | Test Acc: 0.8565  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 24/30 | Train Loss: 0.1144 | Test Loss: 0.1981 | Test Acc: 0.8580  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 25/30 | Train Loss: 0.1183 | Test Loss: 0.1960 | Test Acc: 0.8607  <--
[NEW BEST TEST ACCURACY SAVED]
Epoch 26/30 | Train Loss: 0.1144 | Test Loss: 0.1971 | Test Acc: 0.8559
Epoch 27/30 | Train Loss: 0.1124 | Test Loss: 0.1992 | Test Acc: 0.8574
Epoch 28/30 | Train Loss: 0.1125 | Test Loss: 0.1993 | Test Acc: 0.8565
Epoch 29/30 | Train Loss: 0.1130 | Test Loss: 0.1994 | Test Acc: 0.8574
Epoch 30/30 | Train Loss: 0.1106 | Test Loss: 0.1996 | Test Acc: 0.8577

>>> Finished! Saved Bi-LSTM from Epoch 25 (Acc: 0.8607) as
'BiLSTM_final_model.pth'

Final Classification Report for Bi-LSTM:
                  precision    recall  f1-score   support

Non-Personal (0)       0.91      0.89      0.90      2364
    Personal (1)       0.74      0.79      0.77       967

        accuracy                           0.86      3331
       macro avg       0.83      0.84      0.83      3331
    weighted avg       0.86      0.86      0.86      3331
```


True Test Accuracy: LSTM vs Bi-LSTM
LSTM Test Acc
Bi-LSTM Test Acc
Accuracy
Epochs
True Test Loss: LSTM vs Bi-LSTM
LSTM Test Loss
Bi-LSTM Test Loss
Focal Loss
Epochs


Test Set Confusion Matrix (Standard LSTM)
Actual Label
Non-Personal (0)
Personal (1)
2099
265
233
734
Non-Personal (0)
Personal (1)
Predicted Label
2000
1750
1500
1250
1000
750
500
250




```
>>> All plots and predictions successfully saved!
```